

Step 1: The Data

The RCTA dataset - the first year of IARPA's Hybrid Forecasting Competition - consists of three core files included in a single data release, linked primarily by `discover.question.id`. The questions-and-answers file (`rct-a-questions-answers.csv`) contains 955 rows in total and lists 343 distinct forecasting questions -averaging approximately 2.8 answer options per question- referred to as Intelligence Forecasting Problems (IFPs), along with their resolution outcomes. Each row corresponds to a specific answer option for a given question, with the dataset distinguishing between 184 non-ordinal questions and 159 ordinally scored questions (each with multiple ordered categories). The prediction-sets file (`rct-a-prediction-sets.csv`) contains all participant-level probability forecasts submitted by individual participants during the competition. Each entry records a single forecaster's probability estimate for a particular answer option at a specific time; forecasters are identified by anonymised membership GUIDs. This file includes approximately 750,000 entries, encompassing both initial submissions and subsequent revisions. The daily-forecasts file (`rct-a-daily-forecasts.csv`) captures a daily snapshot of each competing method, recording the most recent probability estimate submitted before 2:01 pm Eastern Time on each scoring day. With over 9.2 million rows, the file contains 320 distinct methods and represents the aggregated outputs of both automated and hybrid performer systems. The aggregates presented below are derived from the individual prediction-sets file, with information on question scope and resolution dates used in the accuracy weighting step sourced from the questions-and-answers file. The daily-forecasts file is included for completeness but does not serve as a source for the analysis.

Step 2: Data Cleaning

Before aggregation, the individual forecast data were filtered as follows. First, the analysis was limited to single-answer binary questions - those with only two possible outcomes and a single focal proposition- which allows for direct application of the standard Brier scoring formula. Ordinally scored questions (159) and non-ordinal questions with multiple answer options (18), both of which require distribution-level aggregation and a more complex cumulative scoring method, were accordingly excluded, leaving 166 single-answer questions. Second, only the focal answer option (`answer.sort.order = 0`) was retained. Third, forecasts submitted after the correct answer became publicly known were removed, as these do not represent genuine predictions. Fourth, records with missing forecast probabilities or missing resolution outcomes were omitted. Fifth, forecasts assigned an absolute probability of zero or one were replaced with 0.001 and 0.999 to satisfy the requirements of geometric mean calculations. A single zero-valued forecast would otherwise reduce the entire

geometric product to zero, nullifying the influence of all other contributors. This adjustment was also required for the geometric mean of odds, for which probabilities of exactly 0 or 1 produce zero or infinite odds, respectively. In total, 16,054 forecasts were adjusted from 0 to 0.001 and 3,161 from 1 to 0.999.

After these procedures, the dataset contained exactly 138,560 individual forecasts from 4,518 unique forecasters, covering 88 binary questions with resolved outcomes. A subsequent deduplication process addressed 994 rows in which a forecaster submitted conflicting probability estimates for the same question at precisely the same timestamp- a raw-data artefact spanning 470 duplicate groups. One row was retained per group - the higher probability value- removing 524 rows in total and leaving 138,036 forecasts. For each question on each active day, a forecaster's contribution was defined as their most recent probability submission up to that point; forecasts submitted early and left unrevised were carried forward to all subsequent days the question remained active. The resulting dataset included 6,667 question-day pairs, and given the carry-forward structure, every question-day pair in the final dataset had a minimum of three contributing forecasters. Thus, every aggregate reflects at least a minimum group estimate rather than a single individual's view.

Step 3: Aggregations and Brier Scores

All five aggregation methods were evaluated using the Brier score, which measures forecasting accuracy by squaring the difference between the probability assigned by a method and the actual outcome, where the outcome is 1 if the event occurred and 0 if it did not. Lower Brier scores indicate better performance: a perfect forecaster scores 0, while a method that always assigns a 50% probability would score 0.25. The table below presents the mean Brier score for each method, averaged across all 6,667 question-day pairs in the dataset. Questions active over longer periods therefore contribute more observations to the mean than short-lived questions, consistent with standard tournament scoring practice. The geometric mean outperformed all other methods, while the raw mean performed worst. This pattern reflects a characteristic of the dataset: only 12.6% of question-days resolved to a positive outcome, indicating that most forecasted geopolitical events did not occur. Forecasters recognised this, assigning low probabilities to the focal answer option on average, with a median individual forecast of 0.157 and the distribution heavily concentrated near zero. In this context, methods that pull group estimates toward lower probability values are rewarded. The geometric mean achieves this most strongly, as it is always equal to or below the arithmetic average, compressing the aggregate toward the low-probability

territory where most correct answers are found. The geometric mean of odds is theoretically appealing because it treats 10% and 90% as symmetrically extreme, which is appropriate in balanced settings. However, when the distribution of outcomes and forecasts is as skewed as in this dataset, the direct downward compression of the geometric mean is better calibrated. The median and trimmed mean provide some resistance to extreme forecasts but do not consistently shrink estimates as much as the geometric mean, explaining their intermediate performance. The raw mean performs worst because it is most susceptible to high-probability outliers, which pull the group estimate upward, away from the low-probability region where accuracy is rewarded.

Rank	Method	Mean Brier Score
1	Geometric Mean	0.0705
2	Geom. Mean of Odds	0.0851
3	Median	0.0972
4	Trimmed Mean	0.1011
5	Raw Mean	0.1104

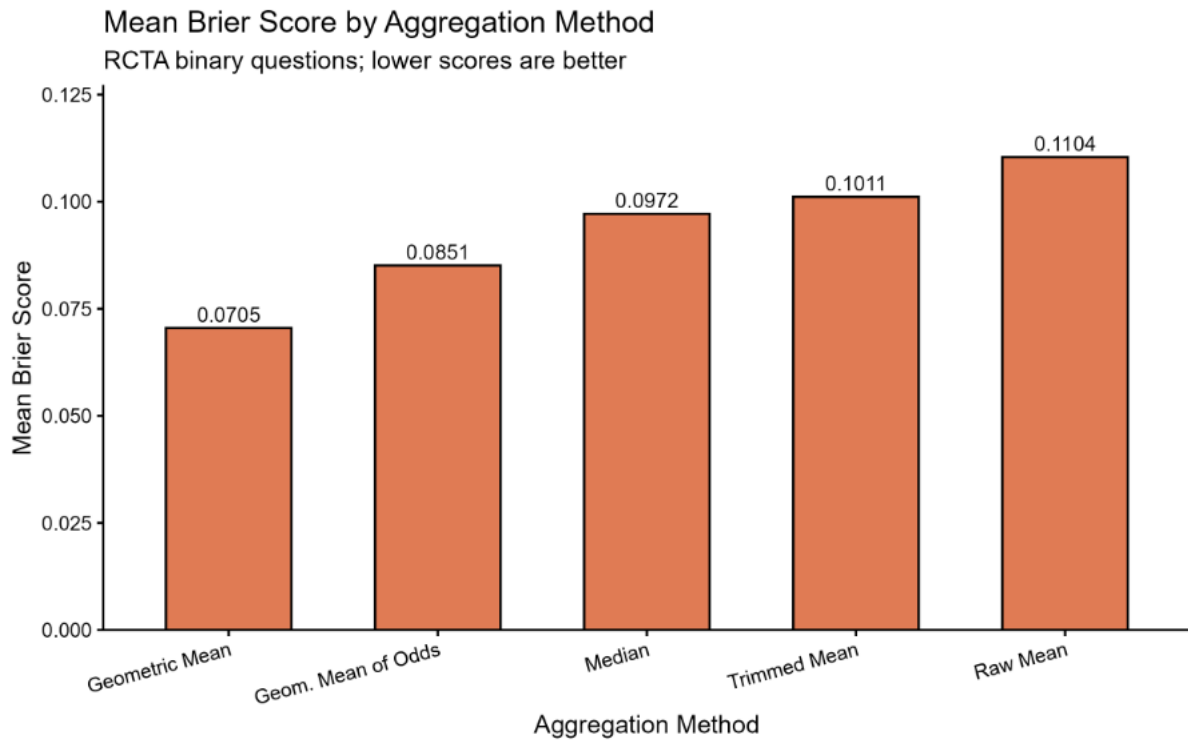


Figure 1. Mean Brier score by aggregation method on RCTA single-answer questions. Lower is better.

Step 4: Improvement

The five aggregation methods discussed above rely on the assumption that all forecasters are equally reliable and that all forecasts are equally up-to-date, thereby not incorporating two properties that jointly determine the quality of the aggregate. The subsequent improvement addresses both omissions. The rationale for this approach is grounded in the empirical observation—well established in the forecasting literature—that individual accuracy exhibits a degree of temporal stability: that is, high-performing forecasters tend to retain their advantage across questions and over time, rather than reverting to the mean (Mellers et al., 2015). In parallel, the carry-forward mechanism results in some forecasts remaining in the dataset for weeks after their initial submission; a probability entered on day one and never revised is likely less informative about current conditions than one updated on the day in question. The improvement implemented here draws on both dimensions. First, an accuracy weight is assigned to each forecaster, calculated as the inverse of their mean Brier score across all questions resolved strictly prior to day T (with a small constant of 0.01 included to avoid division by zero). Forecasters lacking at least five resolved question-days in their record are assigned a default weight of 1, reflecting the absence of sufficient evidence to differentiate their accuracy. Second, a staleness weight is introduced, defined as $1 / (1 + \text{age in days})$, where 'age' refers to the interval (in days) between the forecaster's last

probability update and the current scoring day. A newly submitted forecast receives full weight (1), whereas a ten-day-old forecast receives a weight of approximately 0.09. Both signals are calculated exclusively from data available up to and including day T, ensuring that no information from future periods contaminates the weights. The combined weight for each forecast is simply the product of its accuracy and staleness weights, such that a recent and demonstrably skilful forecast receives the strongest influence, while an outdated and poorly calibrated forecast is downweighted accordingly. This dual-weighted approach generates a mean Brier score of 0.0573, representing a 19% improvement relative to the unweighted geometric mean (0.0705). Disaggregation by resolved outcome indicates that this advantage is most pronounced where the majority of data points occur: on question-days resolving to zero (87.4% of all cases), the mean Brier score falls from 0.034 to 0.025, a 27.6% improvement; for question-days resolving to one, the improvement is 12.2% (from 0.323 to 0.284). Notably, while the accuracy-only approach led to improvements mainly on question-days that resolved to zero, the combined weighting scheme produces substantial gains for both types of outcomes. In sum, the improvement is greatest where most needed, and the two weighting signals function as complements rather than substitutes: skill determines whose input is trusted, while recency determines which of their forecasts remains relevant.

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